# 2.7 Six Digits or Six Hundred

WORKSHEET

**The question**

**STEP II 2023, Question 5**

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2023 STEP papers (a zip file):

www.ocr.org.uk/Images/707102-2023-past-papers.zip

## Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

**Ask yourself** Do you notice anything about the two lists? (Look at which fractions appear in both.)

### STAGE 2 The error in the first recipe

**Ask yourself** The question measures the error by $x_n^2 - 2$ rather than $x_n - \sqrt{2}$. Write $x_{n+1}^2 - 2$ as a single fraction in terms of $x_n$. What do you notice about the numerator?

### STAGE 4 The error in the second recipe

**Ask yourself** Write $y_{n+1} - \sqrt{2}$ as a single fraction. The numerator should remind you of a perfect square.

### STAGE 5 Squaring the error

**Pause and try** Prove by induction that $e_n \le 2r^{2^n}$ for $n \ge 1$, where $r = \frac{\sqrt{2}-1}{2}$. Check the base case carefully.

## What's really going on

#### Newton's method

The second recipe is *Newton's method* applied to $f(y) = y^2 - 2$. Newton's method finds a root of $f$ by repeatedly replacing a guess $y$ by the point where the tangent at $y$ crosses the axis:

$$y_{\text{new}} = y - \frac{f(y)}{f'(y)} = y - \frac{y^2 - 2}{2y} = \frac{y^2 + 2}{2y}.$$

Near a simple root $\alpha$, a Taylor expansion shows that the new error is approximately $\frac{f''(\alpha)}{2f'(\alpha)} \times (\text{old error})^2$, which is why Newton's method converges quadratically. This particular case, "average $y$ and $2/y$", was known to the Babylonians; a clay tablet from around 1700 BC, now called YBC 7289, gives $\sqrt{2}$ in base 60 with an error of less than one part in a million.

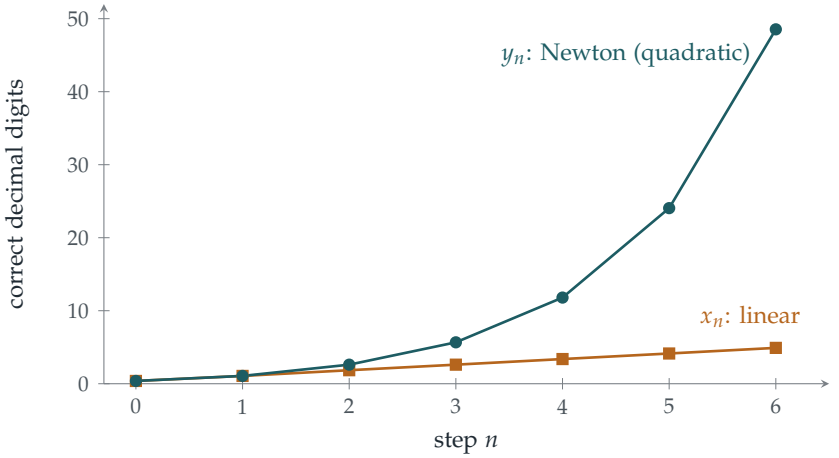



**Figure 2.7.1** The number of correct decimal digits of $\sqrt{2}$ after $n$ steps of each recipe. The linear recipe gains about 0.77 digits per step; Newton's method roughly doubles its number of correct digits at every step, reaching about 780 digits by $n = 10$.

### Fixed points and their speed

Both recipes have the form $u_{n+1} = g(u_n)$ with $g(\sqrt{2}) = \sqrt{2}$, so $\sqrt{2}$ is a *fixed point*, as in the cobweb diagram of Section 2.2. Near a fixed point $\alpha$, the error is multiplied by about $g'(\alpha)$ at each step. For the first recipe, $g(x) = \frac{x+2}{x+1}$ and $g'(\sqrt{2}) = -\frac{1}{(\sqrt{2}+1)^2} = -(3-2\sqrt{2}) \approx -0.17$: the minus sign explains the alternating signs and 0.17 explains the 0.77 digits per step ($\log_{10}(1/0.17) \approx 0.77$). For Newton's method $g'(\alpha) = 0$, so the linear term vanishes and the error is controlled by the next term, the square. Designing iterations with $g'(\alpha) = 0$ is the art of fast numerical methods.

### Why the same fractions appear

The fractions $x_n = p_n/q_n$ satisfy $p_{n+1} = p_n + 2q_n$ and $q_{n+1} = p_n + q_n$, and $p_n^2 - 2q_n^2 = \pm 1$, alternating. These are the solutions of *Pell's equation* $p^2 - 2q^2 = \pm 1$, and they are the convergents of the continued fraction $\sqrt{2} = [1; 2, 2, 2, \ldots]$. They come from powers of $1 + \sqrt{2}$: $p_n + q_n\sqrt{2} = (1+\sqrt{2})^{n+1}$. Newton's step *squares* $p + q\sqrt{2}$ (try it: $(p+q\sqrt{2})^2 = p^2 + 2q^2 + 2pq\sqrt{2}$, and $\frac{p^2+2q^2}{2pq}$ is exactly Newton's formula applied to $p/q$). Squaring doubles the exponent, which is why $y_n = x_{2^n-1}$. The same pattern of "squaring" drives the iteration in Section 2.8.

## Try this next

1 **Pell's equation.** Show that if $y_n = p/q$ in lowest terms with $p^2 - 2q^2 = 1$, then $y_{n+1} = \dfrac{p^2+2q^2}{2pq}$ also satisfies this equation. Check that $577^2 - 2 \times 408^2 = 1$.

**2** **Cube roots.** Write down Newton's method for $f(y) = y^3 - 2$, and show that the new error is $\dfrac{(y_n - \alpha)^2(2y_n + \alpha)}{3y_n^2}$, where $\alpha = \sqrt[3]{2}$. Deduce that if $y_n > \alpha$ then the new error is at most $\frac{1}{\alpha}$ times the square of the old error. (*Hint:* use $\alpha^3 = 2$.)

**3** **Division without dividing.** Show that Newton's method for $f(y) = \frac{1}{y} - a$ gives $y_{n+1} = y_n(2 - ay_n)$, which needs no division. If $e_n = 1 - ay_n$, show that $e_{n+1} = e_n^2$. (Computer processors use iterations like this to divide.)

**4** **When Newton fails.** Apply Newton's method to $f(y) = y^{1/3}$, whose only root is 0. Show that each step doubles the distance from the root, whatever you start with. Which hypothesis of the error formula fails?

## Further reading


- **Endre Süli & David Mayers, *An Introduction to Numerical Analysis*** (Cambridge University Press). The standard undergraduate text; Chapter 1 covers fixed-point iteration and Newton's method.
- **Nick Higham (ed.), *The Princeton Companion to Applied Mathematics*** (Princeton University Press). Short articles on Newton's method and numerical analysis, written by experts for a wide audience.
- **David Fowler & Eleanor Robson, "Square root approximations in Old Babylonian mathematics: YBC 7289 in context"**, *Historia Mathematica* 25 (1998).


The full guided solution is in Section 2.7:
https://beyondstep.keithmcnulty.org/ch02/newton-speed.html